Foreword

Mathematics, Stuyvesant, and Beyond: Change the Question, Continue Searching for Beauty

Professor Joel Lewis, Class of 2003, has been a mathematics professor at George Washington University since 2017 and has completed notable work in enumerative and algebraic combinatorics. While attending Stuyvesant, he was the captain of the Math Team for two years and also an Intel Science Talent Search finalist (now renamed Regeneron). After graduating from Stuyvesant, he attended Harvard College, and graduated with honors in 2007 as a mathematics major. During his college years, he studied at the Budapest Semester in Mathematics program, and he received Honorable Mentions on The William Lowell Putnam Mathematical Competition in 2004, 2005, and 2006. Lewis earned a Ph.D. from MIT in 2012 with his thesis, "Pattern Avoidance for Alternating Permutations and Reading Words of Tableaux," under his advisor, Alexander Postnikov. He then worked for five years as a postdoc at the University of Minnesota and continued his illustrious math career at George Washington University.

0

It is incredibly impressive that Stuy students have been writing publishing a serious math magazine for nearly a century—enough that it was intimidating to be invited to write this foreword. I looked at the contributions of other alumni in recent editions for inspiration (they're at stuymathsurvey.com), but that exacerbated the problem: all those authors are very accomplished, and they've already offered a ton of insight and good advice that is hard to match. But hopefully I can write a few reminisces that are worth reading.

I went to a small middle school and didn't know how I would fit in at Stuy. It was my good fortune that I loved and was very good at solving math problems, and Stuyvesant is one of the best places in the world to go to high school for someone like that. Don't get me wrong, there are other good things about Stuy, too—I fondly remember classes with Ms. Thoms and Mr. Badgley—but if I were the kind of person who used the phrase "spiritual home," I would say that my spiritual home is the 4th floor at 345 Chambers Street, and especially the Stuyvesant Math Team.

0.5

An ant moves along the edges of a tetrahedron. Whenever it reaches a vertex, it chooses one of the three outgoing edges at random. What is the probability that it is back where it started after 17 steps?

What if the ant never walks along the same edge twice in a row?

1

I started as Math Team captain on Tuesday, September 4, 2001. The first thing I remember about that semester is that my schedule had English in the slot when math team met, and Mr. Geller had to go to the program office to get it changed for me. The second thing I remember is that there were more than 100 freshmen on the math team; most of them were in the auditorium, while I had a group of 30 or so in a classroom. The third thing I remember about that semester is being in my (now rescheduled) English class on Tuesday, the second week of the semester; hearing a loud noise from outside; and then later a student running in and telling us that a plane had flown into the World Trade Center.

There's too much about that one day for this short note; suffice it to say that the world changed a lot in the course of a few hours. My GW undergraduates are a few years older than most of you, so I've just recently passed the point where the seminal moments of my life happened before my students were born. Maybe one of you will find yourself writing about going to Stuy via Zoom for the 2042–2043 edition of *Math Survey*—good luck!

We were out of school for—a few days? a week? Then we were relocated to Brooklyn Tech. The Tech students started school in the early morning and finished around noon, then we came in and stayed until the early evening. Periods were a shorter than usual, which was stressful for everyone. In early October, we returned to Stuy. On Chambers Street, we could smell the fires that were still burning underground.

1.5

Suppose that a point (x, y) is to be chosen so that it lies inside or on the boundary of the ellipse $x^2 + 3y^2 = 17$. What is the largest possible value of $3x^2 + y^2$? The smallest?

2

My first experience with "real math" came via the Math Research class. I started working with Joel Spencer, a professor at NYU, during my junior year. Joel (what I call him now—of course back then he was Professor Spencer) gave me some problems to solve (Putnam exam questions and the like), and then I came back the next week and told him my solutions, and then he gave me some more problems to solve, and taught me a little bit about binary search and Ulam's game (see [1]). After a little while of this, he gave me a problem to solve that no one had ever solved before. And that's all math research is, really.

Of course the difference between solving problems designed to be solved in 10 minutes and solving problems whose solutions aren't known is that the second kind of questions are often a lot harder. (Not always: sometimes problems are unsolved just because no one ever asked them before. But often.) So instead of getting a rush of success after a few minutes of struggle, you can spend an hour . . . or a day . . . or a week . . . or a month not making much progress at all.

I remember filling up pages of notebooks with calculations related to Joel's problem, just trying to find a nice pattern. It was hard! In fact, I never solved the problem he gave me—the patterns weren't nice enough for the tools I had available. (It was solved later by Joel and Ioana Dumitriu [2].) Instead, I noticed that if I changed the rules of the game slightly, suddenly there were nice numbers floating around, formulas that could be guessed and proved by induction. Joel told me to forget about the original question and work on the new one.

There may be good life lessons in this story. (Perhaps, not to be afraid of initial failures, and to seek out advice and mentorship.) But for me it is mostly about something I love about abstract mathematics: when you find yourself at a dead end, working with objects that are difficult or ugly, you can always *change the question* and go look for something beautiful instead.

2.5

There is a line that is tangent to the graph $y = x^4 + 2x^3 - 3x^2 + 2x + 1$ at two different points. What are these points?

Which quartic curves have such a "double tangent line"?

3

Let me end with one concrete piece of advice. Many of you will find yourself in college at a place that is even larger and less personal than Stuyvesant. Your professors will have two or three hours a week called office hours. If you are struggling with a class, *go to office hours!* Talk to the professor about the course material, what's giving you trouble, and what to do differently. But, let's be honest: you're Stuy students, so you might not find yourself struggling too often.[1] Suppose you're doing well in your classes; in that case, *go to office hours anyhow!* Tell the professor you're interested in the material and ask for one or two suggestions of something extra to read. Don't be embarrassed or shy; the best part of my job is talking about mathematics with engaged students, and most of my colleagues feel the same way. (The other ones won't remember who you are, so there's not harm in trying.)

3.5

A *swap* in a list takes two adjacent elements and interchanges them, leaving all the others untouched. What is the fewest number of swaps needed to sort the list $(5, 1, 3, 6, 2, 4)$ into its natural order $(1, 2, 3, 4, 5, 6)$?

Suppose I loosen the notion of swap, so you can swap any two entries in the list in one step. (For example, now there are $\binom{6}{2} = 6C2 = 15$ different swaps that can be performed on $(5, 1, 3, 6, 2, 4)$, and one of them produces $(2, 1, 3, 6, 5, 4)$.) How many swaps do I need now?

[1] Let's put the fact that my college freshman-year transcript includes a B−, a C+, and an F somewhere innocuous, like in a footnote. (Needless to say, I did not make good use of office hours in those classes!)

Now suppose the cost of a swap is the distance between the swapped entries (so that swapping the 2 and 5 in $(5, 1, 3, 6, 2, 4)$ has cost 4, because they are 4 spaces apart)—what's the smallest total cost to put it in order?

What if you start with other permutations?

4

Some comments on the questions:

To solve the first question, the first step is to make it harder: set $17 = n$ and try to solve the same question in terms of n. You can probably think of other nice ways to generalize it. If you like this question, you might be interested in learning more about *Markov chains* or the *transfer–matrix method*.

The second and third questions are deceptive. Students in Ms. Pascu's multi class should know one set of techniques for approach the second question—but that's not necessarily the most elegant way to do it. Good luck to solving the third question (which I modified from ARML 2003) with calculus; try to think of an elementary idea.

The last question is near and dear to my heart. The first version is a classic piece of combinatorics that was understood in the 19th century. It is also related to the *bubble sort* algorithm from computer science; the keyword to search for is *inversions*. For the second version, it is helpful to think of a permutation as a function (so $(5, 1, 3, 6, 2, 4)$ represents the function f such that $f(1) = 5$, $f(2) = 1$, $\ldots$) and to consider its *cycles* (like $1 \xrightarrow{f} 5 \xrightarrow{f} 2 \xrightarrow{f} 1$) and to work out what happens to the cycles when you perform a single swap. This was studied by Dénes in the 1950s [3] but may have been known earlier. The third version, which is related to the *straight selection sorting algorithm*, is much newer; it was introduced and solved by Petersen and Tenner in 2015 [4].

Joel Lewis
January 2023

References

[1] *Ulam's game* on Wikipedia. `https://en.wikipedia.org/wiki/Ulam%27s_game`

[2] Dumitriu, Ioana; Spencer, Joel. A Halfliar's game. Algorithmic combinatorial game theory. *Theoret. Comput. Sci.* 313 (2004), no. 3, 353–369.

[3] Dénes, József. The representation of a permutation as the product of a minimal number of transpositions, and its connection with the theory of graphs. *Magyar Tud. Akad. Mat. Kutató Int. Közl.* 4 (1959), 63–71.

[4] Petersen, T. Kyle; Tenner, Bridget Eileen. The depth of a permutation. *J. Comb.* 6 (2015), no. 1-2, 145–178.